EXAMPLE 7.11

Calculate the distance upto which ray optics is a good approximation for light of wavelength 500 nm falls on an aperture of width 0.5 mm.

Solution

$a = 0.5 \text{ mm} = 0.5 \times 10^{-3} \text{ m} = 5 \times 10^{-4} \text{ m}$

$\lambda = 500 \text{ nm} = 500 \times 10^{-9} \text{ m}; \; z = ?$

Equation for Fresnel's distance is, $z = \dfrac{a^2}{2\lambda}$

Substituting,

$$z = \frac{\left(5 \times 10^{-4}\right)^2}{2 \times 500 \times 10^{-9}} = \frac{25 \times 10^{-8}}{1 \times 10^{-6}} = 0.25 \text{ m}$$

$z = 0.25 \text{ m} = 25 \text{ cm}$

7.4.5 Difference between interference and diffraction

It is difficult to find the difference between interference and diffraction as they both exhibit the wave nature of light. In both the phenomena, interference of light only produces maxima and minima on the screen and the diffraction of light only spreads light in the geometrically shadowed region. Nevertheless, in interference, the superposition is given importance and in diffraction, the bending of light is given importance. The difference between interference and diffraction based on the appearance of their patterns are given in Table 7.2.

7.4.6 Diffraction in grating

A grating has multiple slits with equal widths of comparable size to the wavelength of diffracting light. A grating is a plane sheet of transparent material on which opaque rulings are made. A modern commercial grating contains about 6000 lines per centimetre. The transparent space between the rulings act as slit of width a and the rulings act as obstacles having a definite width b. **The combined width of a slit and a ruling is called *grating element* e, ($e = a + b$). The points on the slit separated by a distance equal to the grating element are called *corresponding points*.**

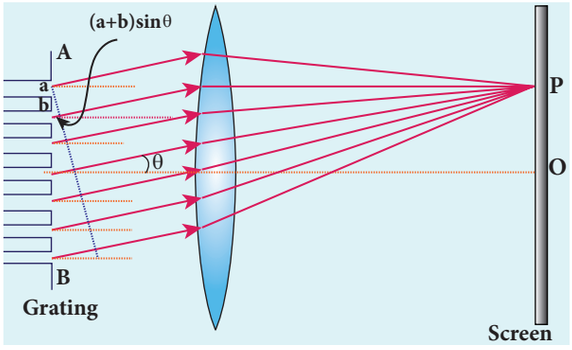


Figure 7.20 Diffraction grating experiment

A plane transmission grating is represented as AB in Figure 7.20. Let, a plane

Table 7.2 Difference between interference and diffraction

S.No.	Interference	Diffraction
1	Equally spaced bright and dark fringes	Central bright is double the size of other fringes
2	Equal intensity for all bright fringes	Intensity falls rapidly for higher order fringes
3	Large number of fringes are obtained	Less number of fringes are obtained

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